

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 10:27 UTC

## SEC P vs NP — notebook summary

Generated 2026-05-18 10:27 UTC

- Cycles recorded: **777**
- Time span: 588.8h (~1.32 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 737   |
| BARRIER_HIT  | 20    |
| FALSIFIED    | 16    |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Real Algebraic Geometry                                                           | 40     |
| Schur-Weyl Duality                                                                | 28     |
| Free Probability Theory                                                           | 19     |
| Representation Theory of Symmetric Groups                                         | 18     |
| Matroid Theory                                                                    | 16     |
| Free Probability                                                                  | 14     |
| Additive Combinatorics                                                            | 13     |
| Noncommutative $L^p$ Geometry                                                     | 10     |
| Random Matrix Theory                                                              | 8      |
| Polymatroid Theory                                                                | 8      |
| Algebraic Geometry                                                                | 7      |
| Spectral Graph Theory                                                             | 7      |
| Finite Geometry                                                                   | 6      |
| Plethysm Theory                                                                   | 6      |
| Persistent Homology                                                               | 5      |
| Noncommutative Harmonic Analysis                                                  | 5      |
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier Analysis on Boolean Functions                                             | 4      |

| Field                                                                                              | Cycles |
|----------------------------------------------------------------------------------------------------|--------|
| Algebraic Geometry over Finite Fields                                                              | 4      |
| Fourier Analysis of Boolean Functions                                                              | 4      |
| Algebraic Geometry of Secant Varieties                                                             | 4      |
| FOURIER_ANALYSIS                                                                                   | 4      |
| Noncommutative Geometry                                                                            | 4      |
| Fourier analysis of boolean functions                                                              | 3      |
| Algebraic Topology                                                                                 | 3      |
| matroid theory                                                                                     | 3      |
| Coarse Geometric Lifting (CGL) — Geometric Complexity Theory                                       | 3      |
| {“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”} | 3      |
| Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)                                              | 3      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                                |
|----------------------|--------------|----------------------------------------------------------------------|
| 2026-05-18 03:29 UTC | INCONCLUSIVE | Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth        |
| 2026-05-18 03:58 UTC | INCONCLUSIVE | Cycle-Type Mixing Defect of Barrington BP Bounds $NC^1$ Length       |
| 2026-05-18 04:25 UTC | INCONCLUSIVE | NW-Design Restriction Dimension Separates Perm from Padded Det       |
| 2026-05-18 04:47 UTC | INCONCLUSIVE | Coxeter Inversion Spread of Barrington BPs Bounds $NC^1$ Depth       |
| 2026-05-18 05:16 UTC | INCONCLUSIVE | Birkhoff Spectral Gap of Gate Co-occurrence Bounds $AC^0$ PARITY Co  |
| 2026-05-18 05:56 UTC | INCONCLUSIVE | Mansour L1 Fourier Mass of Clause-Falsification Polynomial Bounds    |
| 2026-05-18 06:21 UTC | INCONCLUSIVE | Chang Spectrum Dimension of Wire-Source DFS Labels Bounds $ACC^0[2]$ |
| 2026-05-18 06:47 UTC | INCONCLUSIVE | Bourgain Noise-Sensitivity Mass of Clause-Falsification Polynomia    |
| 2026-05-18 07:28 UTC | INCONCLUSIVE | Generated Algebra Dimension of Layer Operators Bounds IP_2 Read-T    |
| 2026-05-18 08:08 UTC | INCONCLUSIVE | Szégedy Quantum-Walk Phase Gap Lower-Bounds Tseitin Tree-Resoluti    |
| 2026-05-18 08:28 UTC | INCONCLUSIVE | Halász $L^2$ Spectrum Discrepancy Lower-Bounds Sign-Matrix Rigidity  |

| Time                 | Verdict      | Title                                                                              |
|----------------------|--------------|------------------------------------------------------------------------------------|
| 2026-05-18 08:56 UTC | INCONCLUSIVE | Mostar Index of Gate-Adjacency Graph Bounds                                        |
| 2026-05-18 09:23 UTC | INCONCLUSIVE | ACC <sup>0</sup> [m] Size for MOD Patience-Sort LIS of XOR-Lifted Row Permutations |
| 2026-05-18 09:54 UTC | INCONCLUSIVE | Bounds CC <sup>D</sup> Effective Resistance of Charged Vertex Pairs Bounds         |
| 2026-05-18 10:26 UTC | INCONCLUSIVE | Tseitin DPLL p=3 Path-Family Modulus Lower-Bounds Tseitin Tree-Resolution          |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```